

PatientTriage.ai: Business Proposal & Enterprise Strategy

“Closing the Emergency Waiting-Room Surveillance Gap”

Accenture Innovation Challenge 2026 | Modeled 500-Bed Hospital ED Impact

Gross Annual Value \$3.82M / yr	Net Annual ROI \$3.58M / yr (14.9x)	LWBS Revenue Recovery +\$1.12M / yr (30% drop)	Target ED Market 5,500+ US/EU EDs
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1. Executive Summary

Emergency departments face a critical surveillance blind spot: triage assesses risk at entry, but patients wait 2.5 to 4.5 hours unmonitored in the lounge. With nationwide nurse turnover at 26.8% and high patient volumes, silent decompensations (sepsis, hypoxia, internal hemorrhage) routinely go undetected until acute collapse. **PatientTriage.ai** is an ambient decision-support copilot designed specifically for the waiting room. By combining continuous vital trajectory tracking, observation shelf-life decay, uncertainty scoring (Unknown ≠ Safe), and clinician attention discounting, the platform delivers a modeled **\$3.82M in gross annual value** (\$3.58M net ROI) per 500-bed facility.

2. Key Stakeholder Value

- Triage Nurses (RNs):** Replaces overwhelming 40+ patient tracking lists with a prioritized 'Next 5 Minutes' action queue, reducing alert fatigue and cognitive strain.
- Emergency Physicians (MDs):** Provides instant visibility into which waiting patients have deteriorated since initial intake, enabling faster, targeted clinical interventions.
- Chief Medical Officers (CMOs):** Minimizes waiting-room sentinel events and builds an append-only audit trail for clinical governance and malpractice defense.
- Chief Financial Officers (CFOs):** Captures high ROI through reduced Left-Without-Being-Seen (LWBS) walkouts, fewer uncompensated ICU transfers, and lower nurse overtime.

Stakeholder	Primary Clinical / Financial Pain Point	PatientTriage.ai Value Proposition
Triage Nurses (RNs)	Overwhelmed tracking 40+ waiting patients; fear of silent deterioration.	Nurse Tasks ('Next 5 Mins') view with 1-click Bedside Reassessment.
Emergency MDs	Blind to which waiting patient has worsened since initial intake.	Attention Gap Queue dispatches doctors to highest unmet clinical need.
Chief Medical Officers	Delayed diagnosis lawsuits, sentinel events in waiting lounges.	100% Downgrade Guardrails & append-only audit trail for malpractice defense.
Hospital CFOs	Uncompensated ICU boarding, LWBS revenue leakage, nurse turnover.	Delivers measurable \$3.58M annual net ROI per 500-bed hospital facility.

3. Financial ROI & Impact Model (500-Bed Facility)

Projections based on 65,000 annual ED visits, 500 beds, and an average \$1,200 baseline ED revenue per visit:

Value Driver	Baseline Scenario	Post-Implementation Target	Annual Financial Impact
1. LWBS Revenue Recovery	3,120 patients walk out (4.8%)	30% reduction via proactive re-engagement	+\$1,123,200 / yr
2. Avoided ICU Escalations	145 waiting-room crashes/yr	64% reduction (93 avoided stays @ \$15k)	+\$1,395,000 / yr
3. Malpractice Risk Mitigation	\$1.2M annual liability allocation	40% reduction via deterministic audit logs	+\$480,000 / yr
4. Staff Retention & Overtime	26.8% RN turnover (14 replacements)	4 turnover events avoided + 15% overtime drop	+\$378,000 / yr
5. Throughput & Boarding	248 min average wait/boarding	30-minute reduction via optimized dispatch	+\$445,000 / yr
Total Gross Annual Value	—	—	\$3,821,200 / yr
Software License & Edge Hardware	—	Enterprise annual subscription + support	-\$240,000 / yr
Net Annual ROI	—	14.9x Return on Investment	+\$3,581,200 / yr

4. Enterprise Architecture & Integration

- **HL7 FHIR & SMART-on-FHIR Native:** Interfaces directly with existing Epic and Cerner systems via standard resources (Encounter, Observation) and CDS Hooks (patient-view), requiring zero rip-and-replace infrastructure changes.
- **Air-Gapped Edge Processing:** Runs locally on hospital servers with sub-15ms inference latency, zero cloud LLM latency dependencies, and no persistent PHI storage in the queue cache.
- **Alarm Fatigue Protection:** Compresses notifications into the top 3 high-yield actions for nurses, preventing alarm flooding while preserving critical escalation paths.

5. Regulatory Stance & Risk Governance

- **FDA CDS Compliance:** Positioned as non-device Clinical Decision Support under 21 U.S.C. § 360aaa-1. Licensed medical professionals retain full decision autonomy and override control.
- **Deterministic Guardrails:** Hard physiological thresholds (SpO2 < 85%, SBP < 75 mmHg) immediately bypass statistical layers to enforce patient safety floors.
- **Fail-Safe Operation:** If telemetry or network connectivity fails, the platform prompts manual rounding intervals with local deterministic alerts active.